

Revision date 18-Jan-2022

Revision Number 1.1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

**Product Name** Precision Plus Protein Dual Color Standards

### Other means of identification

**Catalogue Number(s)** 1610374, 1610374S, 1610394, 10022171, 1610374EDU, 1610374TGX

**Pure substance/mixture** Mixture

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Uses advised against** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
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#### Legal Entity / Contact Address

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ctsthailand@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Singapore: 65-31581349

## SECTION 2: Hazards identification

### GHS Classification

**Serious eye damage/eye irritation**

Category 2

### Label elements



**Signal word**  
Warning

**Hazard statements**

H319 - Causes serious eye irritation

**Precautionary Statements - Prevention**

Wash face, hands and any exposed skin thoroughly after handling

Wear eye/face protection

**Precautionary Statements - Response**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

If eye irritation persists: Get medical advice/attention

**Other hazards which do not result in classification****SECTION 3: Composition/information on ingredients****Substance**

Not applicable

**Mixture**

| Chemical name                                | EC No     | CAS No     | Weight-%     |
|----------------------------------------------|-----------|------------|--------------|
| 1,2,3-Propanetriol                           | 200-289-5 | 56-81-5    | 20 - 35      |
| Sodium lauryl sulfate                        | 205-788-1 | 151-21-3   | 1 - 2.5      |
| 2,3-Butanediol, 1,4-dimercapto-,<br>(R*,R*)- | 222-468-7 | 3483-12-3  | 0.3 - 0.999  |
| Sodium azide                                 | 247-852-1 | 26628-22-8 | 0.01 - 0.099 |

Non-hazardous  
ingredients

Proprietary

Balance

**SECTION 4: First aid measures****Description of first aid measures****General advice**

Show this safety data sheet to the doctor in attendance.

**Inhalation**

Remove to fresh air.

**Eye contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Keep eye wide open while rinsing. Do not rub affected area. Get medical attention if irritation develops and persists.

**Skin contact**

Wash skin with soap and water. In the case of skin irritation or allergic reactions see a physician.

**Ingestion**

Rinse mouth. Never give anything by mouth to an unconscious person. Do NOT induce vomiting. Call a physician.

**Most important symptoms and effects, both acute and delayed****Symptoms**

May cause redness and tearing of the eyes. Burning sensation. Prolonged contact may cause redness and irritation.

**For emergency responders****Self-protection of the first aider**

Avoid contact with skin, eyes or clothing. Wear personal protective clothing (see section 8).

**Indication of any immediate medical attention and special treatment needed**

Note to physicians                      Treat symptomatically.

**SECTION 5: Firefighting measures****Suitable Extinguishing Media**

**Suitable Extinguishing Media**                      Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Unsuitable extinguishing media**                      No information available.

**Specific hazards arising from the chemical**

**Specific hazards arising from the chemical**                      None known.

**Special protective actions for fire-fighters**

**Special protective equipment and precautions for fire-fighters**                      Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

**SECTION 6: Accidental release measures****Personal precautions, protective equipment and emergency procedures**

**Personal precautions**                      Avoid contact with skin, eyes or clothing. Use personal protective equipment as required.

**Other information**                      Refer to protective measures listed in Sections 7 and 8.

**For emergency responders**                      Use personal protection recommended in Section 8.

**Environmental precautions**

**Environmental precautions**                      See Section 12 for additional Ecological Information.

**Methods and material for containment and cleaning up**

**Methods for containment**                      Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up**                      Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards**                      Clean contaminated objects and areas thoroughly observing environmental regulations.

**Reference to other sections**                      See section 8 for more information. See section 13 for more information.

**SECTION 7: Handling and storage****Precautions for safe handling**

**Advice on safe handling**                      Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product.

**General hygiene considerations**                      Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions**

Keep containers tightly closed in a dry, cool and well-ventilated place.

**SECTION 8: Exposure controls/personal protection****Control parameters****Occupational exposure limits**

| Chemical name                 | Singapore                                      | ACGIH TLV                                                                                 |
|-------------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------|
| 1,2,3-Propanetriol<br>56-81-5 | PEL: 10 mg/m <sup>3</sup>                      | No data available                                                                         |
| Sodium azide<br>26628-22-8    | STEL: 0.29 mg/m <sup>3</sup><br>STEL: 0.11 ppm | Ceiling: 0.29 mg/m <sup>3</sup> Sodium azide<br>Ceiling: 0.11 ppm Hydrazoic acid<br>vapor |

**Biological occupational exposure limits**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Appropriate engineering controls****Engineering controls**

Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment****Eye/face protection**

Wear safety glasses with side shields (or goggles).

**Skin and body protection**

Wear suitable protective clothing.

**Hand protection**

Wear suitable gloves.

**Respiratory protection**

No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**Environmental exposure controls**

No information available.

**SECTION 9: Physical and chemical properties****Information on basic physical and chemical properties**

|                 |                          |
|-----------------|--------------------------|
| Physical state  | Liquid                   |
| Appearance      | No information available |
| Colour          | blue                     |
| Odour           | Rotten.                  |
| Odour threshold | No information available |

| <u>Property</u>                        | <u>Values</u>     | <u>Remarks • Method</u> |
|----------------------------------------|-------------------|-------------------------|
| pH                                     |                   |                         |
| Melting point / freezing point         | No data available | None known              |
| Boiling point / boiling range          | 100 °C            |                         |
| Flash point                            | 160 °C            |                         |
| Evaporation rate                       | No data available | None known              |
| Flammability (solid, gas)              | No data available | None known              |
| Flammability Limit in Air              |                   | None known              |
| Upper flammability or explosive limits | No data available |                         |
| Lower flammability or explosive limits | No data available |                         |

|                           |                          |            |
|---------------------------|--------------------------|------------|
| Vapour pressure           | No data available        | None known |
| Vapour density            | No data available        | None known |
| Relative density          | No data available        | None known |
| Water solubility          | Miscible in water        |            |
| Solubility(ies)           | No data available        | None known |
| Partition coefficient     | No data available        | None known |
| Autoignition temperature  | 248 °C                   |            |
| Decomposition temperature |                          | None known |
| Kinematic viscosity       | No data available        | None known |
| Dynamic viscosity         | No data available        | None known |
| <u>Other information</u>  | No information available |            |

## SECTION 10: Stability and reactivity

### Reactivity

Reactivity No information available.

### Chemical stability

Stability Stable under normal conditions.

### Explosion data

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

Possibility of hazardous reactions None under normal processing.

### Conditions to avoid

Conditions to avoid None known based on information supplied.

### Incompatible materials

Incompatible materials None known based on information supplied.

### Hazardous decomposition products

Hazardous decomposition products None known based on information supplied.

## SECTION 11: Toxicological information

### Information on likely routes of exposure

#### Product Information

|              |                                                                                                                                                                          |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. May cause irritation of respiratory tract.                                                             |
| Eye contact  | Specific test data for the substance or mixture is not available. Causes serious eye irritation. (based on components). May cause redness, itching, and pain.            |
| Skin contact | Specific test data for the substance or mixture is not available. May cause irritation. Prolonged contact may cause redness and irritation. Causes mild skin irritation. |
| Ingestion    | Specific test data for the substance or mixture is not available. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea.                        |

### Symptoms related to the physical, chemical and toxicological characteristics

**Symptoms** May cause redness and tearing of the eyes. Prolonged contact may cause redness and irritation.

### Acute toxicity

#### Numerical measures of toxicity

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (oral) 24,275.00 mg/kg  
ATEmix (inhalation-dust/mist) 48.80 mg/l

### Component Information

| Chemical name                                  | Oral LD50             | Dermal LD50                                 | Inhalation LC50                      |
|------------------------------------------------|-----------------------|---------------------------------------------|--------------------------------------|
| Water                                          | > 90 mL/kg ( Rat )    |                                             |                                      |
| 1,2,3-Propanetriol                             | = 12600 mg/kg ( Rat ) | > 10 g/kg ( Rabbit )                        | > 570 mg/m <sup>3</sup> ( Rat ) 1 h  |
| Sodium lauryl sulfate                          | = 1288 mg/kg ( Rat )  | = 200 mg/kg ( Rabbit )                      | > 3900 mg/m <sup>3</sup> ( Rat ) 1 h |
| 1,3-Propanediol,<br>2-amino-2-(hydroxymethyl)- | = 5900 mg/kg ( Rat )  |                                             |                                      |
| Sodium azide                                   | = 27 mg/kg ( Rat )    | = 20 mg/kg ( Rabbit )<br>= 50 mg/kg ( Rat ) |                                      |

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Skin corrosion/irritation** May cause skin irritation. Classification based on data available for ingredients.

**Serious eye damage/eye irritation** Classification based on data available for ingredients. Causes serious eye irritation.

**Respiratory or skin sensitization** Based on available data, the classification criteria are not met.

**Germ cell mutagenicity** Based on available data, the classification criteria are not met.

**Carcinogenicity** Based on available data, the classification criteria are not met.

**Reproductive toxicity** Based on available data, the classification criteria are not met.

**STOT - single exposure** Based on available data, the classification criteria are not met.

**STOT - repeated exposure** Based on available data, the classification criteria are not met.

**Aspiration hazard** Classification not possible.

## **SECTION 12: Ecological information**

### Ecotoxicity

**Ecotoxicity** Harmful to aquatic life with long lasting effects.

**Unknown aquatic toxicity** Contains 0 % of components with unknown hazards to the aquatic environment

| Chemical name         | Algae/aquatic plants | Fish                                         | Crustacea                           |
|-----------------------|----------------------|----------------------------------------------|-------------------------------------|
| 1,2,3-Propanetriol    | -                    | LC50: 51 - 57mL/L (96h, Oncorhynchus mykiss) | EC50: >500mg/L (24h, Daphnia magna) |
| Sodium lauryl sulfate | EC50: =53mg/L (72h,  | LC50: 15 - 18.9mg/L (96h,                    | EC50: =1.8mg/L (48h, Daphnia        |

|              |                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|              | Desmodesmus subspicatus)<br>EC50: 30 - 100mg/L (96h,<br>Desmodesmus subspicatus)<br>EC50: =117mg/L (96h,<br>Pseudokirchneriella<br>subcapitata)<br>EC50: 3.59 - 15.6mg/L (96h,<br>Pseudokirchneriella<br>subcapitata) | Pimephales promelas)<br>LC50: 8 - 12.5mg/L (96h,<br>Pimephales promelas)<br>LC50: 22.1 - 22.8mg/L (96h,<br>Pimephales promelas)<br>LC50: 4.3 - 8.5mg/L (96h,<br>Oncorhynchus mykiss)<br>LC50: =4.62mg/L (96h,<br>Oncorhynchus mykiss)<br>LC50: =4.2mg/L (96h,<br>Oncorhynchus mykiss)<br>LC50: 10.2 - 22.5mg/L (96h,<br>Pimephales promelas)<br>LC50: 6.2 - 9.6mg/L (96h,<br>Pimephales promelas)<br>LC50: 13.5 - 18.3mg/L (96h,<br>Poecilia reticulata)<br>LC50: 10.8 - 16.6mg/L (96h,<br>Poecilia reticulata)<br>LC50: =1.31mg/L (96h,<br>Cyprinus carpio)<br>LC50: =7.97mg/L (96h,<br>Brachydanio rerio)<br>LC50: 9.9 - 20.1mg/L (96h,<br>Brachydanio rerio)<br>LC50: 4.06 - 5.75mg/L (96h,<br>Lepomis macrochirus)<br>LC50: 4.2 - 4.8mg/L (96h,<br>Lepomis macrochirus)<br>LC50: =4.5mg/L (96h, Lepomis<br>macrochirus)<br>LC50: 5.8 - 7.5mg/L (96h,<br>Pimephales promelas) | magna) |
| Sodium azide | -                                                                                                                                                                                                                     | LC50: =0.7mg/L (96h, Lepomis<br>macrochirus)<br>LC50: =0.8mg/L (96h,<br>Oncorhynchus mykiss)<br>LC50: =5.46mg/L (96h,<br>Pimephales promelas)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | -      |

**Persistence and degradability**

**Persistence and degradability** No information available.

**Bioaccumulative potential**

**Bioaccumulation** There is no data for this product.

| Chemical name         | Partition coefficient |
|-----------------------|-----------------------|
| 1,2,3-Propanetriol    | -1.76                 |
| Sodium lauryl sulfate | 1.6                   |

**Mobility**

**Mobility in soil** No information available.

**PBT and vPvB assessment**

| Chemical name         | PBT and vPvB assessment         |
|-----------------------|---------------------------------|
| 1,2,3-Propanetriol    | The substance is not PBT / vPvB |
| Sodium lauryl sulfate | The substance is not PBT / vPvB |
| Sodium azide          | PBT assessment does not apply   |

Other adverse effects

Other adverse effects No information available

**SECTION 13: Disposal considerations**Waste treatment methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

**SECTION 14: Transport information**

ADR Not regulated

IMDG Not regulated  
**Transport in bulk according to Annex II of MARPOL and the IBC Code** No information available

IATA Not regulated

**SECTION 15: Regulatory information**Safety, health and environmental regulations/legislation specific for the substance or mixtureSingapore**Environmental Protection and Management (Hazardous Substances) Regulations**

Verify that licence requirements are met.

| Chemical name         | Hazardous Substances                                                                                                                                                                                                                                                                                                                                                                                                                       | transport |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Sodium lauryl sulfate | Exclusions: 1. Preparations containing <5% by weight of anionic surface active agents. 2. Preparations containing anionic surface active agents which are not <90% biodegradable under a test carried out in accordance with that part of the OECD method which is referred to as Confirmatory Test Procedure in European Communities Council Directive No. 73/405/EEC or other equivalent test methods acceptable to the Director-General |           |
| Sodium azide          | Exclusions: Air bag devices in motor vehicles                                                                                                                                                                                                                                                                                                                                                                                              | 0kg       |

**Environmental Public Health Act**

Dispose of waste product or used containers according to local regulations.

**Hazardous Waste (Control of Export, Import and Transit) Act**

It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste identification and disposal methods in compliance with applicable regulations. Take note that wastes

may be subject to export, import, or transit controls pursuant to the Basel convention and/or local regulations implementing the Basel convention.

**Poison**

None Listed

**Workplace Safety and Health Act**

See section 8 for national exposure control parameters. Comply with the health and safety at work laws.

**International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**SECTION 16: Other information****Key or legend to abbreviations and acronyms used in the safety data sheet**

Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AELG(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Label elements**

P264 - Wash face, hands and any exposed skin thoroughly after handling  
P273 - Avoid release to the environment  
P337 + P313 - If eye irritation persists: Get medical advice/attention  
P501 - Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Issuing Date**

Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 18-Jan-2022**Revision Note** Reviewed existing information and made minor updates.**This safety data sheet complies with the requirements of: SS586: 2008 (2014)****Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**